

Yunjin (Vivian) Yu

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Education

University of California, San Diego

Expected Graduation: June 2027

B.A. - Interdisciplinary Computing and the Arts (Music), minor in Electrical Engineering

- GPA 3.9/4.0
- Coursework: Digital Signal Processing, Machine Learning for Music, Music Programming, Circuits and Systems

Experiences

Sonic Arts R&D - Undergraduate Researcher

July 2026 - Present

Qualcomm Institute, under Professor Shahrokh Yadegari

- Built and optimized preprocessing that auto-oriens SOFA files and maps their measurements onto a consistent coordinate system before applying HRIR truncation, improving the interpolation pipeline's runtime efficiency by ~6x.
- Developed a generic KNN triangle-mesh builder to enable interpolation across any SOFA files.

Projects

Daisy Seed Theremin [in progress]

July 2026 - Present

<https://github.com/vivianyunjinyu/theramen>

- Building a real-time, contactless synthesizer on Daisy Seed (STM32H750) in C++ (libDaisy/DaisySP) that mapped two VL53L0X ToF sensors to continuous pitch and volume over a shared I2C bus, reassigning their identical default addresses at startup via XSHUT sequencing, alongside a 5-position waveform switch.
- Applied one-pole low-pass filtering for glide and decoupled the interrupt-driven audio callback from sensor polling.

Vocoduh - FFT Channel Vocoder

May 2026

<https://github.com/vivianyunjinyu/FFT-Vocoder-Plugin>

- Built a real-time FFT channel vocoder plugin (C++/JUCE) with STFT analysis/resynthesis engine, sidechain carrier routing, and a custom animated GUI with reported latency compensation and parameter handling via AudioProcessorValueTreeState.
- Implemented the vocoder DSP from scratch: overlap-add STFT with Kaiser windowing, mel-scale band grouping, per-band attack/release envelope smoothing, carrier spectral whitening, and frequency-weighted noise injection for unvoiced-consonant intelligibility.

Whoosh - Tempo-Synced LFO Tool

June 2026

<https://github.com/vivianyunjinyu/Whoosh-Plugin>

- Built a tempo-synced volume-LFO plugin (C++/JUCE) that recreates sidechain pumping without an external input by phase-locking ducking curves to the host transport via PPQ position (re-anchored every processBlock), featuring a free-running fallback clock and rate divisions up to 1/16.
- Implemented a draggable two-breakpoint curve editor driving modulation via a 10-bit lookup table with linear interpolation that supports exponential, logarithmic, and linear recovery curves with dirty-flag regeneration.

Skills

Programming: Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib), C, C++, Java, MATLAB, Simulink

Tools & Formats: Git/GitHub, LTSpice, Onshape, PureData

Lab Equipment: Keysight Oscilloscope, Multimeter, Power Supply, Waveform Generator, Soldering Tools

Audio/Video: AVID Pro Tools (Certified Specialist), Ableton Live, Adobe Premiere, DaVinci Resolve

General: Microsoft Office (Word, Excel, PowerPoint), Google Workspace, Canva